Zk-Synergy: Interoperable Blockchain Privacy

The growing demand for multi-organization applications, such as supply chain management, requires coordination between multiple organizations to share and process data. For example, a logistics firm, a pharmaceutical distributor, and a medical institution possess individual blockchain systems. Organizations have to communicate and synchronize information across organizational boundaries, sharing transaction information but keeping sensitive information, for example, prices and supply chains, as a secret. It is not easy to maintain inter-organizational communication and synchronization due to having different consensus protocols and blockchain networks. Therefore, a secure and efficient cross-organization interaction process ensuring privacy is essential.

Blockchain technology has emerged as a potential solution, as it enables data to be stored in a decentralized manner without relying on a central authority. This makes information immutable, verifiable, and not possessed by any entity. For this reason, most digital systems are turning to blockchain-oriented solutions instead of conventional centralized databases. While blockchain gives a sense of security, transparency, and data integrity, ensuring privacy in cross-organization transactions is still a major challenge.

Zk-Synergy, our proposed system, addresses four key challenges in multi-enterprise private blockchain networks. First, it addresses the cross-organization interoperability problem between private blockchains. Second, it solves consensus as there are several consensus mechanisms between networks. Third, synchronization is attained to enable simultaneous transactions. Lastly, privacy must be upheld in data exchange between organizations.

To ensure cross-organization privacy, Zk-Synergy integrates zero-knowledge proofs (ZKP). Given the computational cost of ZKP proof generation, this task is performed off-chain, while verification is executed on-chain for efficiency, preventing overload of the blockchain framework. Within each organization, departments are modeled as shards, and transaction latency is analyzed for both intra-shard (within a department) and inter-shard (across departments in different organizations) operations. Our results show that transaction type and the number of shards significantly affect latency. To improve performance, cross-organization transaction batching is applied, reducing latency for high-frequency operations. The system is also tested with invalid transactions to assess robustness under realistic conditions. The framework employs the Groth16 proving system with multiple elliptic curves, including BLS12-377, BLS12-381, BN254, and BW6-761. The circuit has 3,556 constraints with one public and one secret input for balance verification. BN254 is chosen for evaluation due to its smallest proof size (577 bytes) and compact proving key (1.2 MB), minimizing storage and communication overhead. Batch processing, optimized proof pipelines, and the Gnark framework ensure scalability, efficiency, and privacy. Overall, our framework delivers secure, efficient, and privacy-preserving cross-organization blockchain transactions.

References:

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